Predicting Singapore HDB Rents with Comparable Transactions and Temporally Diverse Ensembles

Han Linxuan A0350911M
Sun Shenghe A0354849L
Zhang Yuhong A0356795H
Zhang Zhexuan A0355290A

***Abstract*—We study monthly rental prediction for Singapore Housing and Development Board (HDB) flats using 150,000 labelled transactions. Market change and ambiguous flat identities limit comparisons with historical rents. We evaluate market-normalised comparable statistics and training-history diversity within an eight-component CatBoost ensemble. In local validation on January–March 2025, the selected ensemble achieves root mean squared error (RMSE) 440.76 and mean absolute error (MAE) 316.55 SGD. Temporal comparisons show gains in recent and longer-horizon windows but regression in an older annual window. Transactions at or below 2,500 SGD contribute 52.23% of recent squared error despite representing only 16.34% of observations. Further model and information tests do not establish a stable replacement. These findings support comparable-market information while identifying limits to its transfer across periods.**

***Index Terms*—HDB rental prediction, tabular regression, comparable transactions, temporal validation, ensemble learning**

I. Introduction

For prospective tenants comparing accommodation costs and landlords assessing comparable rents, useful estimates must account for both flat characteristics and changing market conditions. We estimate Singapore Housing and Development Board (HDB) monthly rent from observable attributes and surroundings, and examine when historical comparables remain informative.

The task uses HDB rental approval information and auxiliary datasets. We treat it as dated cross-sectional transaction prediction: repeated observations of individual flats may exist, but the released fields do not establish their identities. Several flats in a block can share their type, model and area. Block-level histories therefore describe comparable transactions, not verified trajectories of individual flats.

Market-normalised comparables describe local premiums relative to flat-type markets. We combine them with CatBoost models trained on normal and deliberately stale history, then test whether recent gains transfer across periods.

The contribution is empirical. *RQ1* evaluates consistent preprocessing and spatial joins (Sections II and V); *RQ2* evaluates comparable-market estimation and history diversity (Sections III–V); *RQ3* examines temporal transfer and the groups driving residual errors (Section V). Local chronological evaluation is primary; competition scores are supplementary. Code and cumulative evidence accompany the analysis.[1]

CS5228 Group 44 (ELEPHANT_in_the_Data), National University of Singapore.

[1]https://github.com/ElephantLikeCode/cs5228-hdb-rental-prediction

II. Data Understanding and Preparation

A. Released files and task boundaries

The training file contains 150,000 rows from January 2021 through March 2025; the unlabelled competition file contains 50,000 rows from March 2025 through July 2026. Ten inputs describe approval month, town, block, street, flat type, flat model, floor area, furnishing status, lease commencement year and fee; training additionally supplies monthly rent.

March 2025 contains 3,032 training and 380 competition rows, so month precision does not establish strict separation for every transaction. Competition rent labels are not supplied; models and submissions use no public-source test-period targets. Approval month need not equal negotiation month.

Training rent has mean 2,719.04, median 2,700, standard deviation 749.73, and range 300–7,600 Singapore dollars (SGD) (Fig. 1). No core input is missing, although complete fields do not establish correctness or complete economic information. Auxiliary tables describe blocks, Mass Rapid Transit (MRT) stations, malls, primary schools, Certificate of Entitlement (COE) prices and stock prices. A normalised block lookup covers every released rental row without missing coordinates or principal block attributes.

B. Cleaning guided by exploratory analysis

We lowercase categories, trim whitespace, replace punctuation with spaces and collapse repeated spaces. For example, `3-room` and `3 room` become the same category. Raw flat-type labels reduce from eleven to six and street labels from 1,146 to 576 in training. Identical deterministic rules apply to training and competition inputs. This prevents formatting variants from fragmenting comparable groups while preserving immutable raw CSVs.

Furnishing status and fee are constant and are dropped. Calendar year, month, quarter, a month index and nonnegative lease age are derived. Floor area is in square metres; comparable-area keys round it to 0.1 square metres. Lease age uses the recorded lease commencement year rather than renovation condition or tenancy duration. Address combines normalised block and street. Block joins also use town and enforce many-to-one matching, preventing accidental row multiplication.

CatBoost receives canonical strings explicitly identified as categorical columns. Linear and Ridge baselines use training-fitted one-hot encoding with unknown values ignored, median

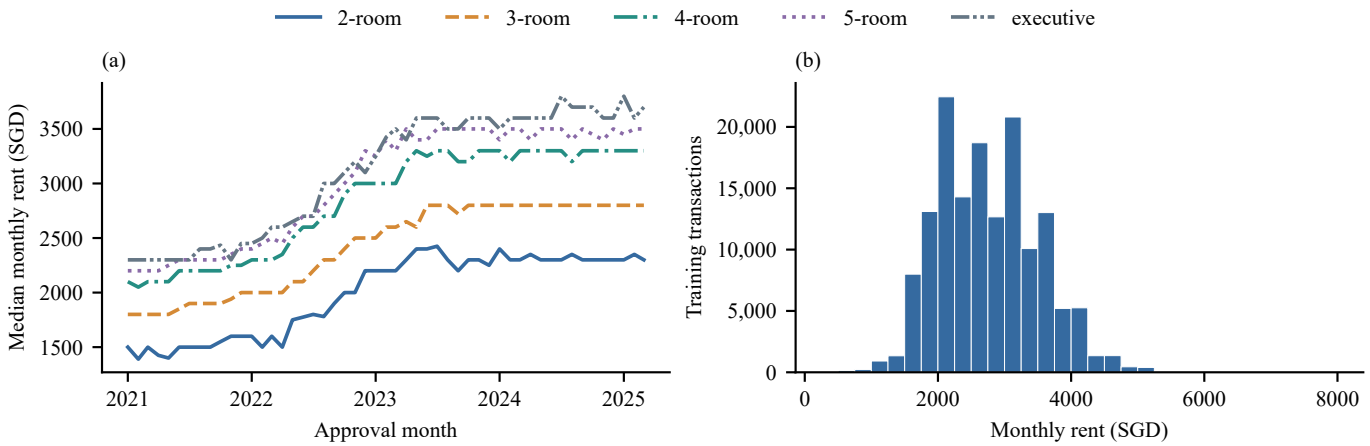


Fig. 1. Recorded rents vary by flat type and approval month. Left: monthly median rent by canonical flat type; one-room flats are omitted because there are only 42 training observations. Right: all 150,000 training rents. Curves describe changing transaction samples, not repeat-flat or causal price indices.

imputation and numeric scaling. The scikit-learn tree baselines use training-fitted ordinal encoding, an unknown-category sentinel and numeric median imputation. XGBoost uses native categorical splits with vocabularies fitted on training only. Raw core inputs are complete; median imputation is retained as a training-fitted fallback for missing numeric inputs. These representations and model capacities differ, so the baseline comparison does not isolate algorithm choice alone.

Fig. 1 motivates two subsequent choices. Monthly medians change markedly, making time-invariant group averages inadequate as the only predictor. Flat-type differences also persist, motivating market normalisation within flat type rather than pooling all room counts into one index. Neither descriptive pattern identifies a causal effect, because transaction composition changes over time.

C. Repeated inputs and extreme rents

An initial audit finds 425 exactly repeated labelled rows. Equal recorded fields and rent are not sufficient evidence of the same flat or contract, so these observations remain. Normalised feature vectors can also repeat with substantially different rents. Removing such records would change the frequency of transaction types without resolving absent individual-flat identity.

All 13,758 recurrent building/type groups have constant supplied area, model and lease year, although historical resales allow multiple areas within a group. This supports caution about unit identity, without recovering the course's attribute-join rule or a particular tenancy's floor.

Low rents for larger flats motivate a sensitivity study. A fixed rule flags three-room-or-larger or executive observations below half the median for the same approval month, town and canonical flat type. It flags 954 training rows, or 0.636%. Such a target-defined flag is a diagnostic or training-policy intervention; it cannot be computed from an unlabelled test transaction. We evaluate alternative training policies on the same complete local validation population and preserve every competition row in its original order. We neither label all low rents as errors nor multiply them by a presumed room count.

III. Methods

The pipeline standardises transaction inputs, joins spatial information, constructs past-only comparable rents and averages eight CatBoost components (Fig. 2). Let x denote a transaction's derived feature vector and y its recorded monthly rent.

A. Spatial information and baseline learners

Block features include coordinates, maximum floor, completion year, subzone, planning area and region. Great-circle distances connect each block to the nearest MRT station, mall and primary school; counts within 0.5, 1 and 2 km describe nearby amenities. These are geographic proxies rather than walking or transit travel times. Predictive association with proximity must not be described as a causal premium.

Global and town/flat-type medians establish simple references. Linear Regression, Ridge, Decision Tree, Random Forest [1] and histogram gradient boosting span increasingly flexible approaches. CatBoost [2] handles numeric and high-cardinality categorical inputs and is selected empirically. Later, matched-input XGBoost [3] tests learner complementarity. These comparisons do not establish a universally preferred learner.

B. Past-only comparable statistics

For query month t, histories contain only permitted training transactions strictly before t, excluding all query-month rents. For nonempty group g, the smoothed mean is

$$\widehat{\mu}_{g,t} = \frac{\sum_{i \in H_{g,t}} y_i + \lambda \mu_t}{|H_{g,t}| + \lambda}, \quad \lambda = 20. \tag{1}$$

In (1), $H_{g,t}$ contains earlier transactions matching key g, i indexes their rents y_i, and $|H_{g,t}|$ counts them. Prior μ_t averages all permitted earlier training rents across flat types;

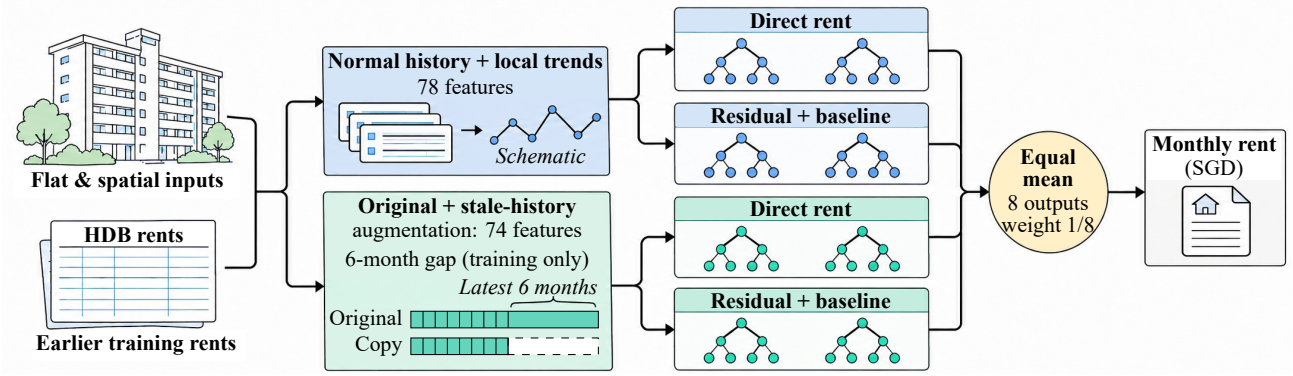


Fig. 2. Selected model: two history views × two target modes × two seeds give eight components. Residual predictions restore their comparable-rent offsets before averaging. Augmentation preserves transaction weight; inference in both views uses permitted earlier training rents, never scoring-interval labels.

λ controls smoothing. Keys are town/type, address/type, address/type/rounded area, street/type and town/type/model, plus address, town, type and model individually. They identify comparables, not unique flats. Unmatched groups retain missing means and zero log support. Separately, adaptive levels use a fixed 2,000 SGD cold-start prior.

Unadjusted rents confound local premium with market movement. The adaptive representation divides each reference rent by its own flat-type/month mean. Decayed premiums follow the parent hierarchy town/type, town/street/type, address/type, and address/type/area. Month indices use $m = 12\,\text{year} + \text{month}$. Weights $w_i = 2^{(m_i - a)/12}$ anchor at the latest permitted reference month a, giving

$$\widehat{r}_{g,t} = \frac{\sum_{i \in H_{g,t}} w_i r_i + \alpha_g \widehat{r}_{p(g),t}}{\sum_{i \in H_{g,t}} w_i + \alpha_g}. \tag{2}$$

Here r_i is the rent ratio, $p(g)$ the parent group, and α_g its smoothing strength: 30, 12, 8 and 5 for the four groups, respectively. The root parent premium is one. Both m_i and a use that month index; weights halve every twelve months. Reference anchoring prevents support vanishing solely with query distance. Each group supplies its market-adjusted level, premium, log support mass and observation age.

For flat type f, $M_{f,a}$ averages its latest three observed monthly means, weighted by transaction counts. Fitting log monthly mean rent against month index over up to twelve observed months gives slope β_f, set to zero with fewer than six and clipped to $[-0.01, 0.01]$ per month. For horizon $h = t - a$ months from the final reference month,

$$\widehat{M}_{f,t} = M_{f,a} \exp\left[\beta_f \, 6(1 - e^{-h/6})\right]. \tag{3}$$

The six-month damping scale makes the effective horizon $6(1 - e^{-h/6})$ approach six, rather than stopping predictions after six months. Multiplying (3) by the premium in (2) gives a comparable-market level. Damping limits extrapolation, not regime risk; its coefficients are design choices, not fitted to outer validation labels.

C. Local trends and training-history diversity

Four town/flat-type features capture movement relative to the flat-type market: normalised premium in the latest six reference months, premium in the preceding six, their log ratio clipped to $[-0.15, 0.15]$, and log support based on the smaller period count. Both period ratios shrink toward one with strength 20. Thus the method measures local relative momentum rather than fitting an unconstrained future trend independently for each block.

A second training view simulates stale information. Eligible observations are copied with the latest six months of reference labels hidden, while preserving their actual query dates. Eligibility requires at least twelve months since the first reference month. The original and copy each receive half the observation's original weight, preserving total weight. This creates alternative information conditions, not new independent rental evidence. At inference, all components use the full permitted training history for each query.

Each view has a direct rent model f and a model g for residual $y - b_v(x)$, where $b_v(x)$ is the adaptive address/type/area level. Averaging both target modes over seeds 42 and 2026 yields

$$\widehat{y}(x) = \frac{1}{8} \sum_{v} \sum_{s} \left[f_{v,s}(x) + b_v(x) + g_{v,s}(x) \right], \quad v \in \{\text{normal}, \text{stale}\}, \quad s \in \{42, 2026\}. \tag{4}$$

The normal view includes local trends; the augmented view omits those four columns, giving input dimensions of 78 and 74. Equation (4) averages the restored predictions without fitting another layer to the latest evaluation errors. Fig. 2 shows the component count; "head" below denotes a view/target pair averaged over two seeds.

D. Alternative hypotheses

Stacking [4] uses predictions generated by earlier training origins to fit a second layer. We compare four seed-averaged heads with all eight individual outputs, using constrained weighting or Ridge correction. Meta-training always precedes meta-evaluation. The retained probe fits nonnegative sum-to-one four-head weights on October–December 2024 and shrinks

them halfway toward equal weights; January–March 2025 labels are not used to fit those weights.

Nearest comparables use up to 64 same-type transactions, scaled location/area/lease-age distances, inverse-squared-distance weights and eighteen-month recency decay. Market-normalised rents are restored using an origin-frozen flat-type market level. Flat-type specialists cover one/two-room, three-room, four-room and five-room/executive groups, with fallback below 500 training rows. Rent-band experts cover training rents $\leq 2{,}500$, $(2{,}500, 4{,}000]$ and $> 4{,}000$ SGD; a classifier supplies prediction-time probabilities. Each enters a fixed 25% blend with the selected ensemble.

Five-fold cross-fitted statistics exclude each inner fold's labels; out-of-fold (OOF) residual corrections learn from earlier-origin predictions. An isolated area/age composition-trend ablation preserves original history levels and residual offsets, unlike an earlier joint change.

Economic inputs comprise lagged supplied COE/stock summaries and release-lagged Consumer Price Index (CPI) and Urban Redevelopment Authority (URA) private-rental indices, with missingness and staleness indicators. An audited MRT calendar supplies operational access, opening exposure and interchange/line-prefix proxies; these are geographic associations, not routes or causal effects. Resale features describe trailing two-year area, floor and price distributions from 2017–2024 records [5], with a two-month lag and frozen validation origin. Building-stock features use a current property snapshot [6], not reconstructed historical stock.

A separate matched-input CPU screen tests LightGBM [7] and an embedding multilayer perceptron (MLP). Log-target CatBoost fits $\log y$ and restores currency predictions by Duan's smearing method [8]: exponentiated predictions are multiplied by the mean exponentiated training residual, without validation-label calibration.

IV. Experimental Design

A. Local validation and metrics

Chronological validation reflects the predominantly later competition period without assuming a pure time series. Each origin fits on earlier months and retains the complete scoring population. Models and rental-label histories remain fixed throughout its validation interval: no validation rents update later predictions. The recent quarter uses 141,076 training and 8,924 validation rows. Fig. 3 shows the local evaluation spans. Table I records cutoffs for all four windows. These overlapping windows have been repeatedly inspected during development and are not untouched final tests.

Root mean squared error (RMSE) emphasises large deviations, mean absolute error (MAE) measures typical absolute error, and signed bias distinguishes systematic overprediction from underprediction. Error is prediction minus observation. For subgroup g, squared-error share is $\sum_{i \in g}(\widehat{y}_i - y_i)^2 / \sum_i (\widehat{y}_i - y_i)^2$. Counts accompany subgroup scores: high RMSE in a rare group can contribute less total error than moderate RMSE in a common group.

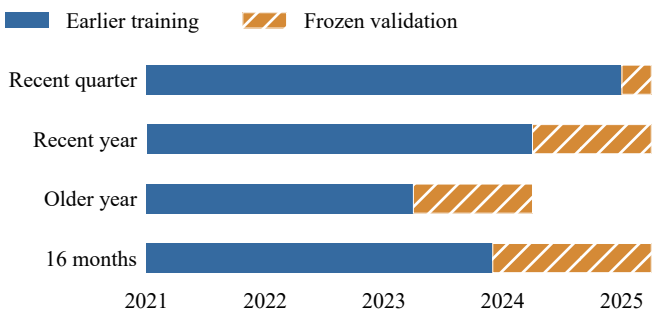


Fig. 3. Four local development windows. Each row fits on earlier months and scores the orange interval without updating rental-label histories. Windows overlap; they are not independent untouched tests. Exact cutoffs and sample sizes are in Table I.

Exploratory CatBoost references use scoring-window early stopping with patience 100. Controlled comparisons instead fix 1,600 trees and supply no outer labels for early stopping. Changes across these protocols cannot be attributed solely to features; neither uses hidden competition labels.

B. Budgets, promotion and uncertainty

Selected CatBoost settings are 1,600 trees, learning rate 0.04, depth 8 and leaf L_2 regularisation 6, shared by deployment and fixed-budget validation. Alternative depths, sample weighting and targets are bounded checks, not exhaustive tuning; historical settings remain in the evidence ledger.

History shrinkage, the twelve-month half-life and trend damping are pragmatic choices held fixed subsequently, without independent optimisation. Four- and eight-component ensembles have equal per-component budgets but different total compute, so diversity and model count are not isolated.

Later screens usually start with seed 42 on the recent quarter and sixteen-month window. Candidates replace a matched direct/residual pair in the frozen ensemble or enter a fixed 25% blend, not eight-model retraining. Further seeds and periods require approximately SGD 2 lower recent-quarter RMSE plus a long-window gain. This practical threshold discourages selecting fitting noise; it is not a statistical significance level.

The CPU CatBoost reference averages four models (two views $\times$ direct/residual targets, seed 42), scoring 441.8429 on the recent quarter; it differs from the eight-model GPU reference. Log-target mixtures must not worsen RMSE on the recent quarter, recent year or older year, and must improve one by at least SGD 2. Boosted trees use 1,600 iterations; the MLP uses 32 epochs. Neither uses scoring-label early stopping. Comparisons use matched references, not a pooled ranking.

Paired comparisons resample address clusters 1,000 times, reporting the 2.5th and 97.5th gain percentiles. Original keys use raw town/block/street strings, potentially splitting buildings. Canonicalised-key sensitivity retains all four gain directions: recent-quarter interval $[3.62, 5.85]$ and older-year $[-11.12, -8.59]$. These exploratory intervals preserve within-key association, not cross-month independence or adjustment

TABLE I
FIXED-BUDGET DEVELOPMENT RMSE (SGD); OVERLAPPING WINDOWS.

Validation interval	Train through	n	Adaptive (4)	Diversity (8)	Local trend (8)
Jan-Mar 2025	2024-12	8,924	445.52	440.89	440.76
Apr 2024-Mar 2025	2024-03	35,158	469.54	466.89	466.76
Apr 2023-Mar 2024	2023-03	35,882	580.54	591.08	590.35
Dec 2023-Mar 2025	2023-11	47,171	486.64	483.83	483.43

Training starts in January 2021 and ends at the listed cutoff. Four or eight components, each with 1,600 trees and seeds 42/2026; total compute differs. No scoring rents enter fitting, early stopping or within-window history updates.

for candidate search. GPU refits can vary; frozen predictions and hashes support auditability, not bit-identical training.

V. RESULTS AND INTERPRETATION

The adaptive reference averages direct and residual models over two seeds (four components); history diversity adds augmented-history counterparts (eight). The selected local-trend model changes only the normal-history half by adding four trend features. Stacking changes their combination weights, not the underlying fitted components.

A. Baseline and spatial information

In Table II, town/type medians outperform the global median, followed by gains from nonlinear trees under the tested representations and budgets. Historical spatial ablations give CatBoost core RMSE 455.71, block 454.19, block plus MRT 452.89, block plus malls 453.97, block plus schools 453.71, and all spatial features 452.62. Correlated address information and imperfect snapshots limit amenity interpretation.

The detailed-comparable two-seed model reaches 448.06 under historical early stopping. Later fixed-budget adaptive direct/residual averaging gives 445.52 on the same recent rows. These milestones use different protocols. Table I supplies the controlled fixed-budget evidence for subsequent selection.

B. Recent gains and an older-period regression

History diversity reduces adaptive-reference RMSE by 4.63 on the recent quarter, 2.65 on the recent year and 2.81 on sixteen months, but worsens the older year by 10.55. Adding local trends further reduces RMSE by 0.13 on the recent quarter, 0.13 on the recent year, 0.40 on sixteen months and 0.74 on the older year. The selected recipe therefore still loses 9.81 versus the adaptive reference on the older year. Its older bias is +82.72 SGD, compared with +8.89 for the adaptive reference. Improvement is conditional on market period (Fig. 4).

The selected recent-quarter RMSE is 440.76, MAE 316.55 and bias +0.79 SGD. Its RMSE gain over the adaptive reference is 4.76, with a 95% paired address-cluster bootstrap interval of $[3.64, 5.87]$. On the older year the gain is -9.81, with interval $[-11.08, -8.59]$. Both use 1,000 resamples of the recorded predictions; these intervals describe the inspected windows rather than an untouched future period. Near-zero aggregate bias also leaves subgroup calibration unresolved.

We retain the selected model because the later-dated task favours recent conditions and it improves both recent and longer-horizon development windows. We accept eight components instead of four for those gains, while retaining the adaptive reference for robustness. Fig. 4 shows the unresolved older-period cost. There, MAE falls slightly from 455.20 to 454.38 while RMSE rises, directing attention to large residuals. Stale-history augmentation changes the training distribution and may overcompensate under a different regime; the observed errors do not establish that mechanism.

Older-year normal/stale RMSE is 579.85/624.08, with bias

TABLE II
EXPLORATORY CORE-INPUT REFERENCES, JANUARY–MARCH 2025 (SGD).

Historical core model	RMSE	MAE
Global median	844.60	691.33
Town/type median	546.81	425.11
Linear regression	523.20	373.09
Ridge	520.01	366.58
Decision tree	485.90	358.59
Random forest	470.19	348.52
Histogram boosting	463.36	336.75
CatBoost (core)	455.71	337.09

CatBoost uses scoring-window early stopping; representations and fitting budgets differ across learners. This is exploratory evidence.

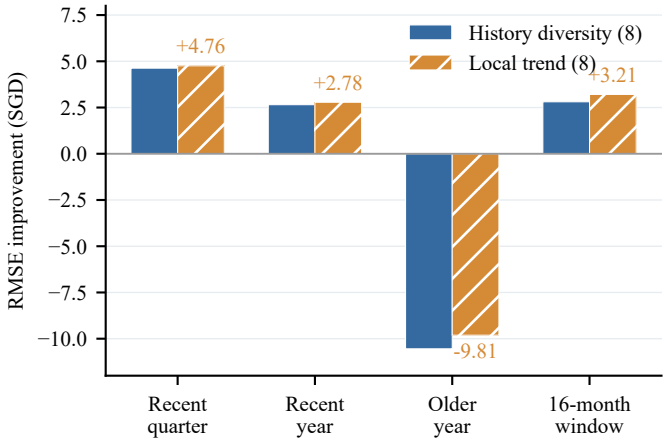


Fig. 4. Recent gains coexist with an older-period regression. Bars show RMSE improvement relative to the four-component adaptive reference; positive is better. They are point differences between fixed ensembles, not seed variability. Windows overlap; paired gain intervals are reported in the text.

+7.08/+158.35; recently, stale history instead scores 441.78 versus normal 445.20. View features differ, so this does not isolate augmentation's causal effect. An earlier-OOF half-bias correction also fails (older/recent RMSE 596.27/440.77).

C. Stacking and supplementary competition evidence

Four-head recent-window stacking improves local recent RMSE from 440.7629 to 440.6565, but its paired gain interval $[-0.0342, 0.2262]$ crosses zero. The eight-output version instead scores 440.8165. Earlier transfer tests are inconsistent; a sixteen-month stress check loses 0.538 with four inputs and 1.189 with eight. That check changes regime, fitting interval and horizon together, so it does not isolate recency weighting. Stacking remains an alternative rather than the default.

Two forward-only weighting studies fit either a horizon-linear normal/stale gate or separate gates for address/type support of 0–4, 5–19 or at least 20 records. Weights shrink toward 50:50. Each improves three of seven overlapping comparisons but worsens the recent quarter (441.08/441.09), recent year and sixteen-month window. These tested gates do not justify replacing equal weights.

Supplementary public RMSE is 432.14881 for the selected model and 432.04286 for stacking. Four-component CPU building-stock/market-normalised probes score 432.70689/434.72053; neither is promoted. Differing compositions and hidden public-subset membership prevent attributing these differences to features. An earlier keep/remove-label probe changes public RMSE by only 0.17 in another model family. It does not establish label errors; the selected model retains all labels and local validation retains every scoring row.

D. Where errors concentrate

Table III and Fig. 5 show that 1,458 rents at or below 2,500 SGD are 16.34% of recent rows but contribute 52.23% of squared error; the corresponding long-window share is 50.86%. Bias is prediction minus observation. This broad band includes plausible low-priced flats and is not an anomaly label. Low observed rents are overpredicted and high rents generally underpredicted. Conditioning on the realised target itself can contribute to this pattern; the plot alone cannot establish omitted contract conditions or incorrect labels as its cause.

Table IV complements outcome-defined bands. Executive flats have recent RMSE 506.73 across 462 rows but contribute only 6.84% of squared error. Three- and four-room groups

TABLE III
RECENT ERRORS BY ACTUAL RENT (SGD); DIAGNOSTIC GROUPS ONLY.

Actual rent	n	RMSE	Bias	SSE (%)
≤2500	1,458	788.1	+589.6	52.2
2501-3000	2,525	294.4	+47.3	12.6
3001-3500	2,823	276.2	-113.2	12.4
3501-4000	1,541	366.7	-252.8	12.0
>4000	577	568.9	-456.0	10.8

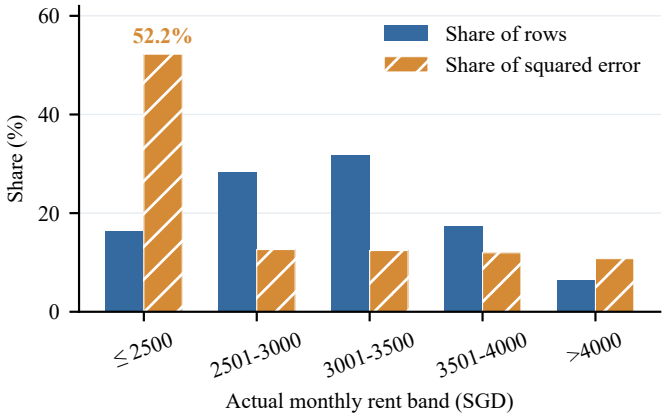


Fig. 5. Share of recent observations versus share of selected-model squared error. The lowest actual-rent band contributes disproportionately. Membership uses observed targets and is unavailable at prediction time.

together contribute 64.87%, partly reflecting prevalence. A narrower diagnostic event, actual rent below 65% of the frozen prediction, occurs in 2.992% of recent rows and contributes 32.47% of squared error. A feature-based detector has area under the receiver operating characteristic curve (AUC) 0.5761 and average precision 0.04161, versus prevalence 0.02992. This weak ranking signal supports neither reliable hard routing nor a claim that these events are completely unpredictable. Both prevalence and error magnitude matter when selecting a specialist population.

For this detector, a depth-4 CatBoost binary classifier uses 600 trees, learning rate 0.04 and $L_2 = 20$. It trains on 35,239 January–December 2024 transactions with earlier-origin OOF predictions, and is evaluated on January–March 2025. Inputs combine observable flat, location and amenity attributes, calendar month and the frozen base prediction; actual query rent only defines the evaluation label. This keeps the diagnostic distinct from the three-band expert classifier.

Housing information is tested against a distinct event: rent below 65% of a strictly-past adaptive address/type/area reference. Its 282 recent rows (3.160%) contribute 32.598% of reference squared error. Added features change AUC from 0.65761 to 0.65835, with gain interval $[-0.00625, 0.00740]$. A two-expert relative-rent mixture and 25% blend score

TABLE IV
RECENT ERRORS BY OBSERVABLE FLAT TYPE (SGD).

Observed flat type	n	RMSE	Bias	SSE (%)
1-room	4	202.5	-27.8	0.0
2-room	242	408.9	+18.2	2.3
3-room	2,915	420.7	+11.9	29.8
4-room	3,201	436.1	0.0	35.1
5-room	2,100	462.8	-12.3	25.9
executive	462	506.7	-13.0	6.8

All 8,924 validation rows; bias is prediction minus observation. Groups are known at prediction time. The four one-room observations are descriptive only; small contributions may round to zero.

453.993/442.223 versus CPU reference 441.843. Some ranking signal exists, but these features do not yield useful routing. Neither event definition identifies verified room rentals.

Historical reliability is assessed from origin-frozen support, recency and dispersion. Dispersion is the interquartile range (IQR) divided by the median of address/type market-normalised rent ratios, requiring at least five records. The high-dispersion group (above 0.25) represents 18.53% of recent rows and 24.88% of squared error. Its RMSE exceeds the low-dispersion group's in all four main windows, including the reported matched strata, but the earliest supplementary annual window reverses the unstratified pattern. Partial stratum coverage and support/recency confounding limit generalisation. Dispersion marks error risk, not verified label errors or a proven smoothing benefit.

Large errors are not confined to rare executive flats. Contract terms, condition and label problems remain unverified explanations; another specialist needs a stable observable error pattern.

E. Exploratory extensions and negative evidence

Table V compares hypotheses under differing budgets. Nearest-comparable, specialist and soft-mixture blends can improve sixteen months while losing recently. Matched-input XGBoost worsens both windows; highly correlated errors limit the tested blend's complementarity. These findings favour the selected recipe within the tested budget.

Cross-fitting excludes a held-out inner fold from training statistics; outer validation uses only outer-training labels. Full-interval statistics worsen both windows, whereas strict-past dispersion replacement gains only about 0.12 and 0.14 RMSE. Changed training/inference support distributions limit generalisation of this finding. The isolated area/age composition-trend ablation provides essentially no recent gain and worsens the other main windows.

TABLE V
EXPLORATORY CANDIDATE COMPARISONS: RMSE (SGD).

Candidate	Use	Recent	16-month
Selected model	–	440.763	483.434
Nearest-comparable blend	B	441.842	482.198
Flat-type specialist blend	B	441.261	483.356
Rent-band mixture blend	B	441.962	482.488
Cross-fitted statistics	R	442.344	483.672
Strict-past distribution	R	440.641	483.299
COE indicators	R	440.877	483.434
Stock indicators	R	440.816	483.536
Matched-input XGBoost	R	441.881	485.244
OOF residual correction	C	440.922	483.340
Guarded Ridge blend	B	442.398	481.515
CPI	R	440.860	483.557
URA rental index	R	440.878	483.528
Corrected static MRT	R	440.631	483.418
MRT opening exposure	R	440.760	483.352
Monthly-available CPI+URA	R	440.791	483.478

B: 25% candidate blend; R: replacement of two selected components; C: 25% OOF residual correction. Candidates use one seed and differing budgets.

a) Parent–child reference accuracy: On identical high-dispersion rows, address/type references outperform address/type/area references by 2.79, 3.20, 4.34 and 3.85 RMSE across the four core windows. The recent gain interval $[-0.65, 6.76]$ crosses zero; the other three exclude zero, but windows overlap. These levels share matched records and the forecast market scale, so their difference reflects smoothing rather than extra area-specific evidence. Street/type references perform worse than address/type in all four windows. Thus the diagnostic motivates a targeted finest-level test, not universally coarser groups. Reference-level gains are not model gains: CatBoost already receives both levels.

b) Dispersion-adaptive shrinkage: For histories with at least five records and dispersion above 0.25, only the finest-level strength in (2) changes from 5 to 10. The corresponding premium/level inputs change; residual targets and prediction offsets retain the original reference. Statistics exclude the query month; validation histories are origin-frozen. Eight CPU fits compare original/candidate direct/residual heads over two windows, seed 42 and 1,600 trees each. Table VI uses an independently executed repetition of the committed recipe, not the full GPU ensemble.

Recent improvement is 0.270 RMSE, but sixteen-month loss is 0.431. Gain intervals from 1,000 normalised address-cluster resamples are $[-0.423, 0.921]$ and $[-0.907, 0.027]$, both crossing zero. Long-window loss occurs in 12/16 months; its first/last eight-month slices worsen by 0.428/0.437. Three-, four-, five-room and executive slices worsen; so does the high-dispersion slice (+0.337). Refitting shared trees can also change predictions for rows whose inputs were unchanged. This fixed intervention does not establish stable benefit.

Model reloads, feature interventions and error totals were verified for the repetition. Provenance and differences from the earlier team run are documented in Appendix A; the two runs support the same non-adoption decision.

COE and stock features use preceding-month summaries and origin-frozen controls. Monthly lags approximate release timing, and stock membership is a supplied snapshot; neither materially improves prediction. Ridge exposes extrapolation instability when future numeric inputs exceed training ranges. Joint range clipping stabilises predictions, but its blend loses recently while gaining over sixteen months. Multiple fields change together, preventing attribution to one variable.

Official CPI [9] activates two months after observation and URA [10] two months after quarter end. Origin-frozen

TABLE VI
INDEPENDENTLY REPEATED SHRINKAGE TEST: TWO-HEAD CPU RMSE (SGD).

Window	Original	Candidate	Δ
Jan–Mar 2025	445.759	445.488	-0.270
Dec 2023–Mar 2025	486.476	486.907	$+0.431$

All 8,924/47,171 validation rows retained. Each prediction averages direct and residual heads. Δ = candidate minus original; negative is better. This is a same-device two-head ablation, not the eight-component GPU comparison.

indices are compared with a same-model control using releases available by each rental month. Revisions and rebasing are not reconstructed; URA private rents are only an HDB proxy. The monthly CPI+URA replacement remains slightly worse than the selected ensemble on both windows, despite improving its long-window frozen-input control.

The MRT audit finds 55 supplied rows marked open outside the verified operational calendar. Research variants verify Thomson–East Coast Line stages 2–4 (TEL2–4), Punggol Coast and Hume dates against Land Transport Authority (LTA) releases (for example, TEL4 [11]), exclude future/unverified codes, deduplicate interchange names, and activate service from the first full month after opening. Origin-based validation admits dated openings announced before that origin; the monthly-access control admits operational service by the rental month. Supplied coordinates limit coverage, especially for light rail transit (LRT). Corrected static rail gains just 0.1322 and 0.0169 RMSE, with intervals crossing zero. Filtering and deduplication change jointly, so their contributions are not isolated. Dynamic access and opening exposure establish neither a reliable improvement nor a causal premium.

The index and rail studies comprise 36 formal GPU fits, excluding smoke checks. An identical-input seed-42 macro refit changes a replacement score by about 0.13; gains of this scale require caution. The selected recipe retains the supplied static MRT snapshot for comparability, not as a claim of operational access at every historical date.

Six additional interchange-access fits use nearest operating-interchange distance and line-code diversity within 1/2 km, admitting only openings announced before each origin. Recent RMSE worsens by 0.084 under replacement and 0.655 under a 25% blend. The latter gains 4.938 on the older year but still loses to the adaptive reference; sixteen-month gain is only 0.100. These geographic proxies provide no stable replacement or causal transport estimate.

Log-target integration averages two log heads with the four-head CPU reference, using one seed and the original feature workflow. Recent/older-year RMSE changes from 441.843/585.844 to 442.120/583.515; recent-year change is 467.304 to 467.272. The recent loss fails the three-window acceptance rule. A separate 91-feature screen reduces sixteen-month bias from +85.21 to +27.98 but also loses recently. Target transformation changes regime bias without uniform benefit; complete results remain in the shared evidence.

The four-head CPU screen scores LightGBM 452.28 and the embedding MLP 471.61 versus reference 441.84. Resale/building-stock replacements give 441.827/441.783, below the required gain. Duplicate-mean, market-normalised, deeper/asymmetric-tree and historical-only augmentation probes also fail advancement. Augmentation adds 4,357 attribute-matched public records [12] strictly before the scoring origin. These bounded negative results do not rule out other representations; latest-vintage inputs and reused windows remain limitations.

Candidate batches stop because gains are small or period-dependent. Further tuning would reuse inspected labels without resolving the older-period failure.

VI. Limitations and Reproducibility

A. Scope of the conclusions

Evaluation reuse. Temporal windows have been inspected repeatedly, several overlap, and public submissions add another selection channel. Selection on a noisy performance estimate can itself overfit [13]; bootstrap intervals do not repair this reuse. We have no private-label evaluation. The older-period regression prevents a claim of general dominance over the adaptive reference.

Incomplete observation. Unit floor, renovation and contract terms are unobserved; flat identity and the course's attribute-join rule remain unresolved. Unusual rents establish neither label errors nor an irreducible-error floor. Predictions estimate recorded transactions, not normative fair rent.

Information availability. Static MRT proxies, revised macro vintages, snapshot stock membership and approval/negotiation timing limit live deployment. Spatial and opening associations are not causal estimates.

Bounded search. One-seed component screens cannot rule out better full retraining, and no individual-tenancy uncertainty interval is supplied. Future work needs stable observable subgroup bias in additional, preferably non-overlapping periods, then confirmation with later data and repeated seeds. Actual commuting routes remain outside the tested geographic proxies.

B. Implementation and reproducibility

Code, configuration and aggregate provenance are shared; immutable course data, fitted models and row predictions remain local. The README gives cache-regeneration commands and Appendix A maps the evidence.

An independent checkout/new Python 3.12 environment repeats eight 1,600-tree fits in 679.83 seconds on the RTX 4060, validating 50,000 predictions and model reloads. Recent refit RMSE is 440.6423 versus frozen 440.7629: fitting variation, not a method gain. Appendix A records dependencies, CPU repetition cost and verification scope; paper compilation needs no private caches.

Public-source target matching is quarantined from modeling. Its target-visible diagnostics are not held-out prediction results and are not used to train, tune or submit models; they establish neither a noise floor nor other teams' methods.

VII. Conclusion

We have shown that market-adjusted comparables and history diversity improve recent HDB rental prediction under the evaluated conditions: RMSE is 440.76 and MAE 316.55, at higher compute than the four-component reference. Gains extend to two longer windows, but the older annual deficit of 9.81 remains. Low rents contribute disproportionate squared error without reliable feature-based detection. Recent accuracy therefore does not establish robustness across market periods. Further work should prioritise new observable evidence over additional tuning of reused windows.

REFERENCES

[1] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[2] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Advances in Neural Information Processing Systems*, vol. 31. Curran Associates, Inc., 2018. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2018/file/14491b756b3a51daac41c24863285549-Paper.pdf

[3] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 2016, pp. 785–794.

[4] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.

[5] Housing and Development Board, "Resale flat prices based on registration date from January 2017 onwards," data.gov.sg, the modeling subset uses 2017–2024 records with time cutoffs. [Online]. Available: https://data.gov.sg/datasets/d_8b84c4ee58e3cfc0ece0d773c8ca6abc/view

[6] Housing and Development Board, "HDB property information," data.gov.sg, current building-stock snapshot retrieved 2 November 2026; historical vintages not reconstructed. [Online]. Available: https://data.gov.sg/datasets/d_17f5382f26140b1fdae0ba2ef6239d2f/view

[7] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[8] N. Duan, "Smearing estimate: A nonparametric retransformation method," *Journal of the American Statistical Association*, vol. 78, no. 383, pp. 605–610, 1983.

[9] Singapore Department of Statistics, "Consumer price index: All items, table M213751," 2024-base historical snapshot downloaded 6 October 2026. [Online]. Available: https://tablebuilder.singstat.gov.sg/table/TS/M213751

[10] Urban Redevelopment Authority, "Private residential property rental index, quarterly," data.gov.sg, non-Landed, Whole Island series. Experimental snapshot downloaded 6 October 2026. [Online]. Available: https://data.gov.sg/datasets/d_8e4c50283fb7052a391dfb746a05c853/view

[11] Land Transport Authority, "TEL4 to welcome commuters from 23 June 2024," 2024. [Online]. Available: https://www.lta.gov.sg/content/ltagov/en/newsroom/2024/3/news-releases/TEL4_to_welcome_commuters.html

[12] Housing and Development Board, "Renting out of flats from January 2021," data.gov.sg, owner-declared rental records; historical-only augmentation and a separate target-visible diagnostic are distinguished in the text. [Online]. Available: https://data.gov.sg/datasets/d_c9f57187485a850908655db0e8cfe651/view

[13] G. C. Cawley and N. L. C. Talbot, "On over-fitting in model selection and subsequent selection bias in performance evaluation," *Journal of Machine Learning Research*, vol. 11, pp. 2079–2107, 2010. [Online]. Available: https://www.jmlr.org/papers/volume11/cawley10a/cawley10a.pdf

APPENDIX A
REPRODUCTION AND EVIDENCE MAP

Obtain the original course files under `data/raw/`, then run `python src/check_setup.py` and `python -m pytest -q`. The README supplies commands to regenerate temporal component predictions before feedback studies. Cached predictions and models are not distributed. Compiling the paper needs committed sources and aggregate assets; rebuilding evidence needs course data and caches. Full-budget selected-model training is verified in an independent checkout and new virtual environment on the same physical machine. GPU refits differ numerically; frozen hashes and metric checks prevent silently replacing old evidence. A new run requires a reviewed manifest and corresponding report update. Cross-device verification is a separate team task.

CatBoost is pinned to 1.2.10. The maintained suite passes 142 tests, with four optional LightGBM tests skipped.

Fixed-budget comparisons use `src/run_experiments.py` with the `--compare-candidates` option. The selected ensemble is generated by `src/train_final_model.py` with the recorded configuration; research uses `src/feedback_experiments.py`, including `--families public` for indices and rail. These are abbreviated entry points; use the complete README commands for exact windows and options.

Supplementary CPU modules are under `src/research/`, with a separate `requirements-research.txt`. Their aggregate evidence is in `reports/november_cpu_comparisons_20261102.csv` and `reports/november_log_integration_20261102.csv`. The consolidated research record preserves protocols and unsuccessful probes.

The shrinkage study uses `src/research/dispersion_shrinkage.py`. Shared repetition results, intervals, attribution and provenance are under `reports/history_research/dispersion_shrinkage_cpu_reproduction_20261106/`. Analysis uses `src/research/analyze_shrinkage_errors.py` with explicit input/output paths and model verification; full commands are in the README. Models and row predictions remain local; original team aggregates are preserved separately.

The CPU shrinkage repetition takes 2,346.23 seconds for fitting/prediction plus 146.67 for feature construction. It reloads all eight models and verifies predictions, feature interventions and error totals. Earlier team-run scores differ by at most 0.199 RMSE across the twelve head/pair rows; unavailable original caches prevent attributing that discrepancy. Both runs support non-adoption, but their numerical results are not interchangeable. For clarity, the historical core-baseline settings are listed in Table VII. They are bounded references, not the outcome of an exhaustive or common-compute search. The selected ensemble's settings and history coefficients are given in the main text. Table VIII maps shared evidence locations.

TABLE VII
PRINCIPAL HISTORICAL BASELINE SETTINGS; PREPROCESSING IS DESCRIBED IN THE MAIN TEXT.

Model	Recorded settings
Linear/Ridge	Ordinary least squares; Ridge $\alpha = 10$
Decision Tree	Maximum depth 18; minimum leaf size 20
Random Forest	180 trees; maximum feature fraction 0.7; minimum leaf size 3
Histogram boosting	250 iterations; learning rate 0.08; 31 leaves; $L_2 = 2$

TABLE VIII
EVIDENCE LOCATIONS; DETAILED LOCAL PREDICTIONS CAN BE REGENERATED.

Evidence	Location
Baseline/spatial results	`reports/experiment_results.csv`
Fixed-budget windows	`reports/temporal_summary.csv`
Candidate comparisons	`reports/feedback_results.csv`
Stacking transfer	`reports/stacking_results.csv`
Actual public scores	`reports/kaggle_submissions.csv`
Selected recipe	`configs/model.json`
Model provenance	`reports/final_model_metadata.json`
Research decisions	`docs/PROGRESS.md`

APPENDIX B
AI TOOL DECLARATION

We used Claude and Codex to discuss research ideas and experimental alternatives, assist with code development, debugging and checks, analyse recorded experimental results, and draft, format and refine the report.

OpenAI's image-generation tool was used to create the schematic model-overview illustration in Fig. 2. Its labels and layout were subsequently refined for inclusion in this report. The numerical plots and tables were generated with code from executed experiments and saved results.

We are responsible for the content and quality of the submitted work.

APPENDIX C
TEAM WORKLOAD STATEMENT

Table IX lists the main responsibilities of each team member.

TABLE IX
MAIN RESPONSIBILITIES OF TEAM MEMBERS.

Member / Student ID	Main responsibilities
Han Linxuan A0350911M	Investigated the reliability of historical comparables, including weighting by forecast horizon and historical support, rent dispersion analysis, parent–child reference comparisons, and targeted shrinkage experiments. Analysed error attribution and temporal transfer, and contributed to the corresponding methodology, experimental analysis, and discussion of limitations.
Sun Shenghe A0354849L	Led the overall project framework and main modeling pipeline, including data cleaning, baseline comparisons, spatial and historical features, CatBoost ensembles, temporal validation, and ablation studies. Conducted model reproduction, testing, and submission-file verification. Took primary responsibility for manuscript drafting, figure preparation, verification of experimental evidence, and final integration of the code and report.
Zhang Yuhong A0356795H	Conducted supplementary model and information-source studies, including LightGBM, MLP, CatBoost target transformations, housing and resale attributes, low-rent detection, and expert mixtures. Consolidated validation results and reasons for retaining or rejecting candidate approaches, and contributed to the corresponding experiments, interpretation of findings, and research documentation.
Zhang Zhexuan A0355290A	Contributed to ensemble-strategy discussions and proposed assigning different weights to the eight CatBoost components, motivating alternatives to equal averaging and informing weighted-fusion and stacking comparisons. Participated in discussions of candidate approaches and experimental results, and reviewed the report, suggesting improvements to methodological explanations, result interpretation, and clarity of presentation.